

Quality Inspection Report QIR-2025-0198

Dimensional / Lot Acceptance Inspection | SLIP, UPPER, SEGMENTED (Cast iron CL 40 w/ HT inserts)

Report No. / Date	QIR-2025-0198 / 2025-06-30
Part Number	ACT-BP-5500-03
Part Description	SLIP, UPPER, SEGMENTED (Cast iron CL 40 w/ HT inserts)
Work Order / PO	WO-25-1502 / PO-88231 (Fort Worth Foundry Co.)
Lot / Heat-Batch	FWF-25-3310 / FWF-C4459
Lot Qty / Sample Qty	240 / 13 (per ACT-QP-08)
Serial Numbers	Lot sample - segments 1-13
Material	Cast iron ASTM A48 Class 40; inserts AISI 52100 HT
Inspected By	D. Whitfield
Reviewed / Approved By	S. Nguyen
Applicable Specs	API 11D1 / API 19V (as applicable), NACE MR0175/ISO 15156, ASTM E709, API 5B

Dimensional & Attribute Results

Actuals are the worst-case (closest-to-limit) reading across the sample unless noted. Instruments are calibrated per ACT-CAL-01; calibration records on file.

Char.	Feature / Characteristic	Nominal	Tol +	Tol -	Actual (worst)	Instrument	A/R
D1	Slip OD (assembled gauge)	4.437	+0	--0.008	4.445	Ring gauge R-55	ACC
D2	Insert protrusion	0.062	+0.010	--0.010	0.072	Depth mic DM-3	ACC
D3	Segment height	3.400	+0.015	--0.015	3.415	Caliper C-21	ACC
D4	Insert hardness (HRC)	55.0	+4.0	-0	51.5	Hardness tester HR-3	REJECT
D5	Body hardness (HRB)	100.0	+7.0	--7.0	107.0	Hardness tester HR-3	ACC
D6	Wicker angle (deg)	60.0	+1.0	--1.0	61.0	Optical comparator OC-1	ACC

Result: REJECT - see NCR

Nonconformance Report NCR-2025-077

NCR Number	NCR-2025-077
Nonconformance	Characteristic D4 (insert hardness): 4 of 13 sampled segments measured 51.5-53.8 HRC, below the 55.0 HRC minimum required by ECO-2025-091 / Dwg D-BP-5500-03 Rev C.
Disposition	REWORK - Lot FWF-25-3310 returned to heat treat for re-austenitize, quench and temper of insert sets. Re-inspection per QIR-2025-0205 (13/13 conforming, lot released 2025-07-15). Root cause: furnace zone 2 thermocouple drift at supplier; supplier CAR SCAR-25-19 issued.
MRB Approval	S. Nguyen

Measuring & Test Equipment Used

Instrument	Cal. Record	Cal. Due	Status
Ring gauge R-55	CAL-4930	2026-07-18	In calibration at time of use
Depth mic DM-3	CAL-5232	2025-10-14	In calibration at time of use
Caliper C-21	CAL-9493	2026-01-04	In calibration at time of use
Hardness tester HR-3	CAL-4462	2025-09-24	In calibration at time of use
Optical comparator OC-1	CAL-4953	2025-01-20	In calibration at time of use

Sampling & Methods

Sampling per ACT-QP-08 (ANSI/ASQ Z1.4 basis). Dimensional measurements at 68-72 degF after minimum 4 hr thermal soak. Surface finish per ASME B46.1. Hardness per ASTM E18 (HRC/HRB) or ASTM E140 conversion where noted. Threads gauged per API 5B with certified working gauges. Magnetic particle inspection, where applied, per ASTM E709 using wet fluorescent method, 100-125 V/in. field.

Inspector Notes

Inspection triggered by ECO-2025-091 implementation audit. Tightened sampling (Level III) applies to next 3 lots from this supplier.

Sign-off

Role	Name	Signature	Date
QC Inspector (Level II)	D. Whitfield	/signed electronically/	2025-06-30
QA Manager	S. Nguyen	/signed electronically/	2025-06-30